iLearn Practice Exam

12BR · Chapter / Topic 18

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Topic 18: Artificial Intelligence

Name: _____

Class: _____

Date: _____

Time allowed: 55 minutes

TOTAL MARKS: 44

Marks obtained: ..... / 44

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 44.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 6 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 44 marks of past-paper questions are available for this chapter, so the recommended time is 55 minutes (1 min 1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 **(a)** Describe the purpose of a graph when used in an Artificial Intelligence (AI) system.

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..... [2]

(b) Explain the use of artificial neural networks in Deep Learning.

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..... [4]

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2 **(a)** Explain what is meant by regression methods in relation to machine learning.

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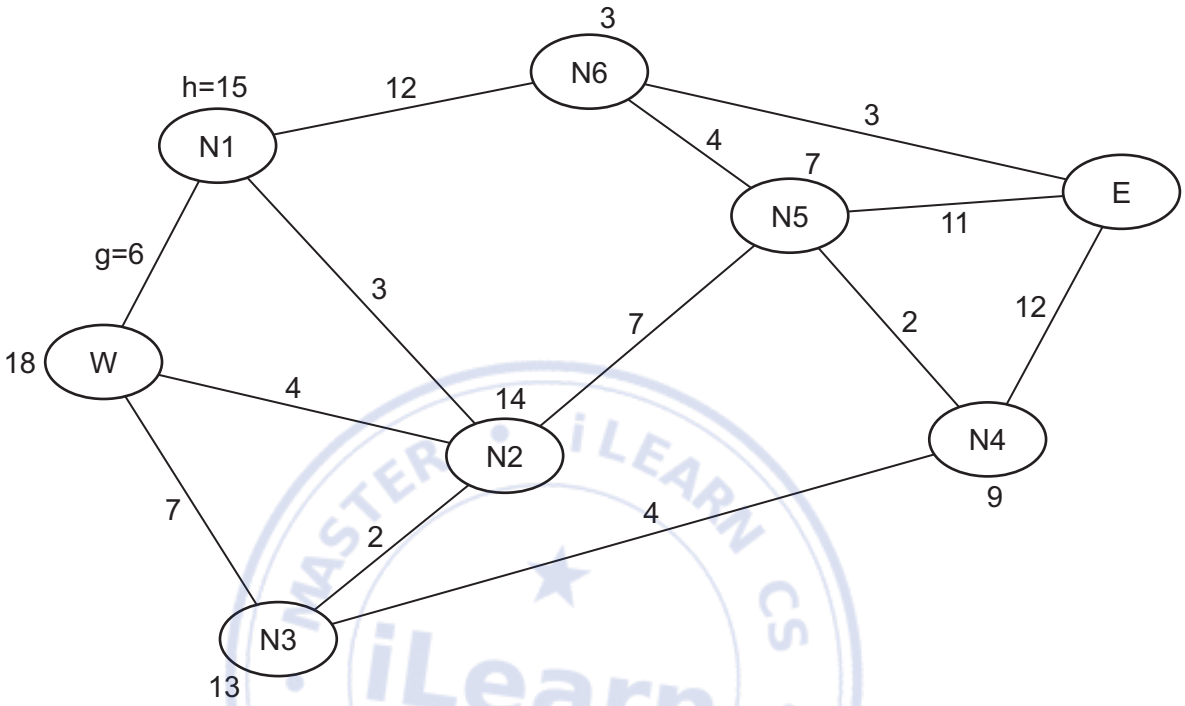
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(b) The A* algorithm is used to perform searches on a graph.

Calculate the best path between the W and E nodes on the following diagram using the A* algorithm.

Show your working in the table provided.
The first two rows have been done for you.



current path	destination node	cost of current path (g)	heuristic (h)	total (f = g + h)
W	W	0	18	18
W	N1	6	15	21

final path	

[5]

3 **(a)** State the subset of Artificial Intelligence (AI) which makes use of an artificial neural network.

..... [1]

(b) The A* and Dijkstra's algorithms are used to perform searches on a graph.

(i) Outline how the A* algorithm is different from Dijkstra's algorithm.

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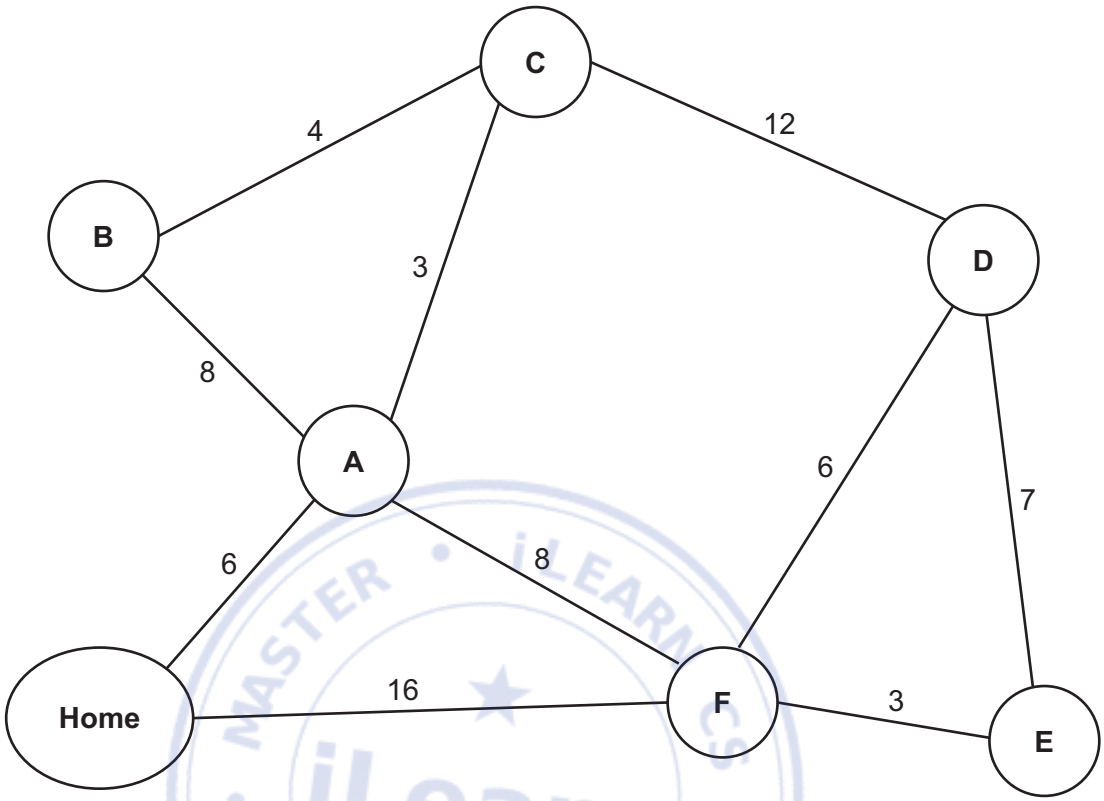
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(ii) Calculate the best path between the Home node and each of the nodes in the diagram using Dijkstra's algorithm.

Show your working and write your answers in the table provided.



working .....

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Answers:

A	B	C	D	E	F

[5]

4 **(a)** Describe the structure of a graph as used in an Artificial Intelligence (AI) system.

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..... [2]

(b) Explain how supervised learning and unsupervised learning differ from each other.

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[4]

5 **(a)** Explain how supervised learning and unsupervised learning are trained.

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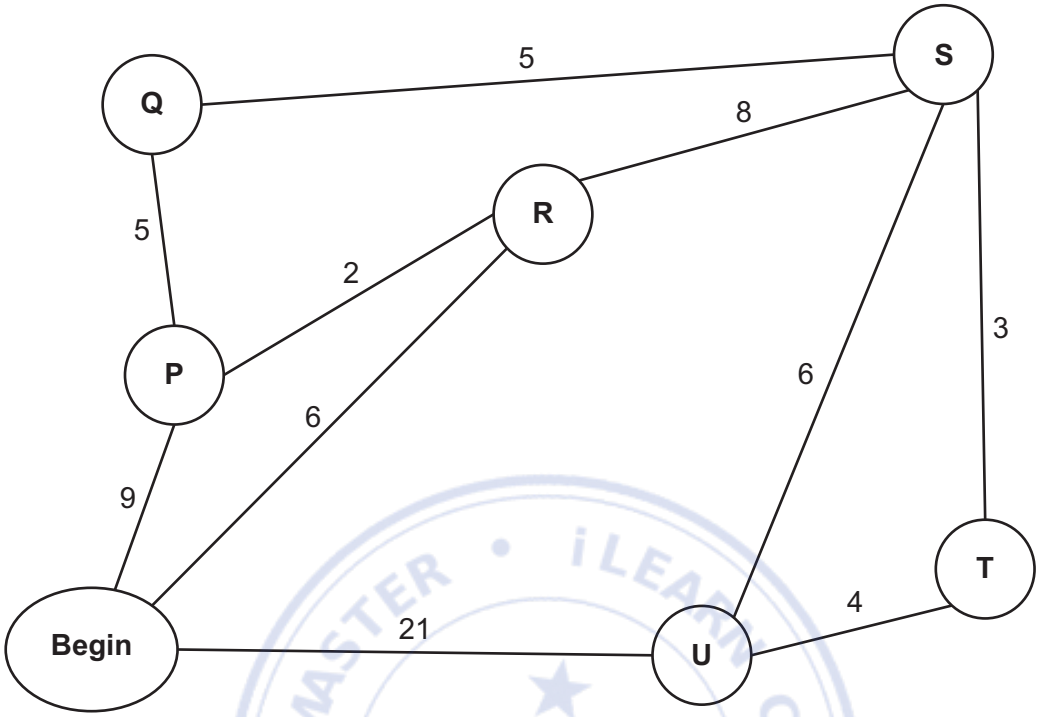
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(b) Calculate the best path between the Begin node and each of the nodes in the diagram using Dijkstra's algorithm.

Show your working and write your answers in the table provided.



working .....

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answers:

P	Q	R	S	T	U

[5]

6 **(a)** State the purpose of the A* and Dijkstra's algorithms.

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..... [1]

(b) Outline the difference between the A* and Dijkstra's algorithms.

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(c) Explain how unsupervised learning takes place in machine learning.

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..... [3]

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END OF QUESTION PAPER

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